

Reading Earth's 4.6-Billion-Year Story

EarthViewer · HHMI BioInteractive

Years 7-10

~30 minutes

Science · Earth & Space Science

NOTICE — AI-generated worksheet

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Simulation: https://media.hhmi.org/biointeractive/earthviewer_web/

Curriculum links: AC9S10U04 · AC9S8U03 · AC9S7U01

Learning intentions

- I can use EarthViewer's timeline to identify major events in Earth's geological and biological history.
- I can describe evidence for plate tectonics shown across deep time.
- I can explain how changes in temperature, oxygen and biodiversity are connected over Earth's history.

Getting started

Open EarthViewer and drag the timeline slider slowly from 4.6 billion years ago towards the present. Turn on the temperature, oxygen and biodiversity data layers using the layer toggles so you can see how they change alongside the continents. Keep the timeline visible as you answer the questions below.

Tasks

1. Drag the slider to 4.6 billion years ago (the very start). Describe what Earth's surface and atmosphere looked like at this point, based on what the simulation shows.

2. Turn on the plate tectonics/continents layer and slowly move the slider forward in time. Describe two changes you notice in the position or shape of the continents.

3. Find a point in the timeline where the continents look very different from how they look today. Explain what evidence the simulation gives you that continents move over time.

4. Turn on the oxygen layer. Find the approximate point in time when atmospheric oxygen levels first rise sharply. What else does the simulation show happening in the biosphere around this time?

5. Choose four different time points on the timeline (spread across Earth's history) and record the temperature and biodiversity trends the simulation shows at each.

Time period	Approx. temperature trend	Approx. biodiversity level

6. Find one of the mass extinction events marked on the timeline. Describe what happens to the biodiversity layer immediately before and after this event.

7. Using your table from Task 5, describe any pattern you notice between temperature and biodiversity across Earth's history. Is a warmer Earth always linked to higher or lower biodiversity in your data?

8. Compare the diversity of life shown near the start of the timeline with the diversity shown today. What does this suggest about how classification systems for living things have needed to change over time?

Extension (optional)

Scientists use evidence layered in rock, ice and fossils to build timelines like this one. Choose one event you explored above (a mass extinction, an oxygen spike, or a continental shift) and describe what kind of physical evidence on Earth today might have been used to figure out when it happened.
